
Kothapalli Ravi Kiran

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My-Portfolio : [Ravi-Kiran-Portfolio](#) | Git-Hub: [Ravi-Kiran-Git-Repo](#)

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Summary

A dynamic professional with a solid foundation in software development and a fresh perspective in the world of data science. With hands-on experience as a **Data analyst** and **Full-Stack-Developer**, I've honed my technical expertise and problem-solving skills. However, my true passion lies in the limitless possibilities of data science.

I recently earned my **MCA in data science** from Jain University, a transformative journey that ignited my enthusiasm for working with data. I'm excited to leverage my development background to uncover insights, patterns, and solutions hidden within data.

SKILLS

Machine Learning: Classification, Regression, K-Means Clustering

Statistical Methods: Predictive Analysis, Text Analytics, Data Mining, Random Forest, Time series Analysis.

Programming Languages: Python, C#, HTML-5, CSS-3, Bootstrap.

Web Scraping: BeautifulSoup

Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Requests.

Database Language: SQL, MS-SQL.

Data Management Tools: JIRA

Data Reporting Tool: Tableau, Power BI

AI Tools: ChatGPT: Proficient in utilizing ChatGPT with effective prompting skills for various purpose and development.

EDUCATION

Jain University, Bengaluru – *Masters of computer applications in Data Science*

Aurora College, Hyderabad - *B. Com Computers.*

Certificates

Exploratory Data Analysis for Machine learning – *IBM.*

EXPERIENCE

Soul AI (Freelancing) – *Prompt Engineer & LLM Trainer.*

September 2025 – Present.

- Joined **Soul AI** as a **Prompt Engineer & LLM Trainer**, responsible for designing and crafting high-quality prompts tailored to specific project needs and guidelines.
- Evaluated **Large Language Model (LLM) responses** for the **Action Extension** project, rating them against established **Rubrics** to ensure quality and alignment with project requirements.
- Analysed Large Language Model (LLM) outputs and refined responses to align with project requirements, contributing to continuous model improvement.
- Adapted prompt design and response analysis strategies across diverse projects to ensure optimal model performance.

Outlier (Freelancing) – *Prompt Engineer & Analyst.*

September 2024 – September 2025.

- I joined Outlier as a Prompt engineer and Analyst here my duties are to design and High-quality prompts according to the project needs and instructions.
- Also, then I used to analysis the response of the LLM's and need to change the response according to the project requirements this will help the model in Learning.
- The prompt Design and Response analysis will be varying from project to project.

Fusion Forte Solutions, Bengaluru – *Data Analyst & Full Stack Developer*

January 2021 – September-2024

- I joined this company as a Full Stack Developer in January 2021, engaging in both front-end and back-end development.
- Subsequently, I expanded my scope to encompass data analysis, report generation, and contributed to projects involving Tableau and Power BI. This experience motivated me to pursue a career in the field of Data Science, leading me to enroll in MCA Data Science.

PROJECTS RELATED TO DATA SCIENCE

1. Predicting Loan Defaults using Machine Learning

The goal of this project was to develop a machine learning model to predict whether a client would default on loan payments. Default prediction is a crucial task in the financial industry as it helps financial institutions assess the risk associated with lending to certain clients. By accurately

predicting loan defaults, banks and other financial institutions can mitigate potential losses and make more informed lending decisions.

Method:

The project utilized supervised machine learning techniques to tackle the classification task of predicting loan defaults. Specifically, a dataset containing various features related to clients' loan applications and financial history was used to train and evaluate the predictive model. Two primary algorithms, Random Forest and XGBoost, were employed to build predictive models due to their effectiveness in handling tabular data and classification tasks.

Project Steps:

1. **Data Collection:** Acquired datasets containing information about loan applicants, including demographic data, financial attributes, and loan terms.
2. **Data Pre-processing:** Conducted comprehensive data pre-processing tasks, including handling missing values, encoding categorical variables, and converting data types for model compatibility.
3. **Feature Engineering:** Performed feature engineering techniques to extract relevant information from raw data and enhance predictive performance.
4. **Model Building:** Utilized Random Forest and XGBoost algorithms to train classification models on the pre-processed dataset.
5. **Model Evaluation:** Evaluated the performance of the trained models using appropriate evaluation metrics such as accuracy, precision, recall, and F1-score.
6. **Hyperparameter Tuning:** Optimized model hyperparameters using techniques such as grid search or random search to improve model performance.
7. **Documentation:** Documented the entire project process, including data pre-processing steps, model development, evaluation results.

Outcome:

The developed machine learning model demonstrated promising results in predicting loan defaults, achieving high accuracy and robust performance across various evaluation metrics. The project showcased the potential of leveraging data-driven approaches to address critical challenges in the financial sector, particularly in risk assessment and decision-making processes related to lending activities.

2.Customer Churn Analysis

- Executed during MCA in Data Science, demonstrating a structured approach to predict customer churn in a business context.
- Conducted comprehensive **Exploratory Data Analysis (EDA)** involving data visualization, statistical analysis, and **feature engineering** to uncover trends and potential factors influencing churn.
- Systematically handled missing values and outliers in data pre-processing, ensuring clean and reliable datasets for analysis.

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- Utilized **one-hot encoding** for effective transformation of categorical variables, enhancing machine learning model performance and interpretability.
 - Implemented machine learning models, including **Random Forest**, **Logistic Regression**, and **XGBoost**, with rigorous tuning and evaluation for optimal performance.
 - XGBoost emerged as the best-performing model, demonstrating superior predictive power in identifying potential customer churn.
 - Employed industry-standard metrics to assess model performance, highlighting technical proficiency and practical relevance in solving real-world business problems.
 - The project's outcomes, particularly the actionable insights derived from XGBoost, showcased my ability to contribute meaningfully to proactive business decision-making.

3.Social Media Tourism

- Developed to transition from traditional tele calling to digital advertising, aiming to optimize marketing efforts through targeted strategies on a social networking platform.
- Implemented a segmentation approach based on user devices, creating distinct predictive models for Laptop and Mobile users to enable device-specific targeting and personalized digital advertisements.
- Executed tailored data analysis and pre-processing for each model, considering the unique characteristics of Laptop and Mobile users, including data cleansing, feature engineering, and outlier detection.
- Employed Random Forest for the Laptop user segment and Gradient Boosting for the Mobile user segment, rigorously optimizing and tuning each model to maximize predictive accuracy.
- Rigorous performance evaluation using precision, recall, F1-score, and ROC AUC metrics confirmed the suitability of Random Forest for Laptop users and Gradient Boosting for Mobile users, showcasing exceptional predictive power.
- The project demonstrated my proficiency in designing and implementing device-specific predictive models, optimizing digital advertising resources, and showcasing a deep understanding of machine learning techniques to cater to the distinct behaviour of users on different devices.

4.Customer Segmentation and Targeted Advertising Optimization

- Executed to enhance the marketing department's effectiveness through data-driven strategies, focusing on market segmentation for targeted ad marketing campaigns within a bank's customer base.
- Leveraged extensive customer data from the past 6 months, employing advanced data science techniques to derive actionable insights for empowering the marketing team.

- **Applied K-means clustering to segment customers** into distinctive groups based on shared characteristics and behaviours, providing a foundation for tailored marketing strategies.
- **Utilized Principal Component Analysis (PCA)** for dimensionality reduction, improving modelling efficiency and enhancing interpretability while retaining essential information.
- The insights gained from K-means clustering and PCA guided the launch of highly effective targeted ad marketing campaigns, improving customer engagement, revenue growth, and overall sales.
- The project showcased my proficiency in advanced data science techniques and their direct impact on optimizing marketing strategies, contributing to business growth and customer satisfaction through distinctive customer segmentation and targeted advertising efforts.

5.Sales Forecasting and Demand Prediction

- Focused on leveraging advanced AI/ML techniques for predicting future daily sales, providing data-driven insights to optimize inventory management, demand forecasting, and drive revenue growth.
- Pre-processed data from 1115 stores, ensuring data quality and completeness, and incorporating historical sales data, seasonal patterns, holidays, and promotional events.
- **Implemented Facebook Prophet**, a state-of-the-art forecasting tool, to develop predictive models capable of handling seasonality effects, trends, and holidays, significantly improving sales forecast accuracy.
- Meticulously trained and evaluated predictive models, emphasizing key performance metrics such as **mean absolute error (MAE)**, **root mean squared error (RMSE)**, and **mean absolute percentage error (MAPE)**, with the chosen model demonstrating exceptional accuracy and reliability.
- Equipped the sales team with actionable insights derived from Facebook Prophet, enabling informed decisions on inventory stocking, marketing campaigns, and sales strategies, thereby enhancing the sales department's capability to meet customer demand effectively.

PROJECTS Related to AI Tools

Rock-Paper-Scissors Game Created with ChatGPT.

- Developed a fully functional Rock-Paper-Scissors game entirely guided by ChatGPT's instructions, **showcasing my proficiency in utilizing ChatGPT's prompting skills for complex application development.**

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- **Leveraged ChatGPT's capabilities to generate code and instructions**, with my role focused on following prompts to implement the suggested code and logic.
 - Highlighted the creative application of AI-driven prompting skills, demonstrating ChatGPT's versatility beyond traditional text-based tasks in producing an interactive game.
 - Designed the game for user engagement, offering an interactive experience guided by ChatGPT's generated content, showcasing my ability to harness AI's creative potential for diverse applications.
 - The project is a testament to my curiosity and experimental approach to AI-driven development, reflecting a commitment to exploring and pushing the boundaries of AI applications.
 - Featured on my resume to convey my expertise in utilizing AI for innovative and creative endeavours, aiming to contribute to the development of unique AI-driven solutions within a dynamic organization.